White Paper on End-Network Collaborative AI Networks for Ultra-Large-Scale AI Training: Joint Architecture Analysis of Asterfusion AI Controller and EasyRoCE

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AI Infrastructure Network Background and Core Challenges

Compared to traditional Internet services, traffic generated by distributed AI Large Language Model (LLM) training (e.g., Pipeline, Tensor, and Data Parallelism) exhibits characteristics of **high throughput, strong burstiness, and extreme sensitivity to packet loss and latency**. In RoCEv2 (RDMA over Converged Ethernet) networks, even minor network congestion or packet loss can trigger global GPU compute stalls, drastically lowering Model Flops Utilization (MFU).
Traditional "device-centric" decentralized management suffers from three primary limitations:

- **Low Configuration Efficiency & High Error Rates:** Manually logging into individual devices via Command Line Interface (CLI) is time-consuming and labor-intensive; human operational error remains a leading root cause of network outages.
- **Complex Configuration Dependent on Empirical Tuning:** Lossless Ethernet involves numerous interdependent parameters—such as Priority-based Flow Control (PFC) watermarks, Explicit Congestion Notification (ECN) thresholds, and Buffer allocations—lacking standardized guidelines and turning parameter tuning into a "black box."
- **Difficult Fault Localization Disrupting Business Continuity:** Fragmented management interfaces, delayed fault detection, and excessively long diagnostic chains, combined with a lack of cross-domain analysis between NICs and switches, result in "Slow Worker" troubleshooting taking days, severely impeding continuous model training.

Core Components and Working Principles of EasyRoCE and AI Controller

Asterfusion's solution constructs an end-to-end, host-network integrated AI networking architecture through deep synergy between the **AI Controller (Management Hub & Cloud-Native Chassis)** and **EasyRoCE (Lossless Optimization Engine & Suite)**.

AI Controller (Management Hub & Cloud-Native Chassis)

The AI Controller operates on the principle of **"Northbound Requirement Intake → Core Microservice Routing & Orchestration → Southbound Gateway Centralized Deployment/Telemetry"**:

- **API Gateway (OpenAPI/RESTful):** Serves as the unified northbound ingress interfacing with Slurm, Kubernetes (K8s), or Web UI requests.
- **Orchestration Service (OWPROV):** Translates computed network strategies into deployable JSON configurations, schedules distribution, and persists state into a PostgreSQL database.

- **Configuration Migration Service (OWMGMT):** Operates independently from routine O&M logic, dedicated to packaging, snapshotting, and idempotently importing the system's full configuration baseline in PostgreSQL.
- **O&M Service Components:** Encompasses `owom` (Alarm and Event Management), `owfms` (Firmware and Patch Management), `owupgrade` (Controller Self-Upgrade), and `owsec` (Security and Access Control).
- **Southbound Gateway (OWGW):** Maintains persistent long-lived connections with massive switch fleets and GPU node agents, responsible for centralized policy push and telemetry aggregation.
- **East-West Message Bus (Kafka) & Analytics Engine (OWANALYTICS):** OWGW writes high-volume telemetry data into Kafka for asynchronous buffering, where OWANALYTICS subscribes in real time for timestamp alignment and multi-source cross-correlation diagnostics.
- **Core Data Storage:** PostgreSQL database, utilized for persistent storage of network policies and runtime states.

EasyRoCE (Lossless Optimization Engine & Suite)

EasyRoCE follows the operating principle of **"Ex-Ante Algorithmic Solving → Host-Side Driver Locking → Physical-Layer Hardware Self-Governance → Multi-Source High-Frequency Telemetry"**:

- **Algorithmic Solving Suite (AID / RPA / ORD):**
  - **AID:** Automatically parses input logical topologies and generates topology blueprints.
  - **RPA:** Combines cluster scale, optical propagation delays, and switch ASIC Buffer specs to precisely solve optimal PFC watermarks and dynamic ECN gradients/thresholds.
  - **ORD:** Integrates AID topology blueprints with RPA mathematical solution results, exposing RESTful / CLI one-click deployment interfaces.
- **Host-Side Management (EasyRoCE-NC):** Automatically parses NIC planning blueprints, executes environment pre-checks, runs standardized configuration scripts in batch, and forcibly locks driver settings on GPU host nodes.
- **Data-Plane Hardware Self-Governance:** When congestion occurs, Leaf switches autonomously trigger microsecond-level PFC hop-by-hop backpressure and ECN end-to-end marking based on ASIC register thresholds preset by the system, ensuring zero packet loss and low latency at the hardware layer without controller intervention.
- **Tripartite Telemetry Collection Components (AE / RE / NE):**
  - **AE (AsterNOS Exporter):** Exports switch CPU, memory, optical transceiver health, and standard network protocol states.
  - **RE (RoCE Exporter):** Exports switch ASIC-level PFC/ECN queue counters and flow control performance metrics.

  - **NE (NIC Exporter):** Exports real-time operational parameters of physical GPU NICs (e.g., RDMA retransmission rates, DMA response latency).

- **Hardware Base & O&M Visualization:** CX-N AI switches running AsterNOS perform adaptive routing and load distribution, paired with EasyRoCE-UG to provide a visual operations dashboard.

Deep Fusion and Two Core Collaborative Mechanisms

Asterfusion breaks the conventional boundary where "network tools only manage switches but remain blind to host NICs," achieving synchronized parameter locking and joint self-healing across both network and host domains.

Mechanism 1: Algorithmic Solving and End-Network Collaborative Deployment Loop

1. **Automated Algorithmic Solving:** When a training job is submitted northbound, AID generates logical topology blueprints while RPA calculates optimal PFC watermarks and dynamic ECN gradients tailored to cluster scale and optical delay.

2. **Policy Orchestration & Persistence:** RPA computation results are transmitted via ORD interfaces to the OWPROV orchestration service and persisted into PostgreSQL.

3. **Synchronized Dual-Path Push:**

   - **Network Path:** OWPROV instructs the southbound gateway OWGW to write PFC/ECN thresholds directly into CX-N switch ASIC registers via `uCentral over WebSocket`.
   - **Host Path:** OWGW simultaneously pushes driver parameters to the EasyRoCE-NC agent on GPU hosts via `Host Agent API`, forcibly locking NIC driver configurations.

4. **Synergistic Value:** Achieves atomic, synchronized parameter locking between network policies and host drivers, eliminating "NIC-Network" misconfigurations.

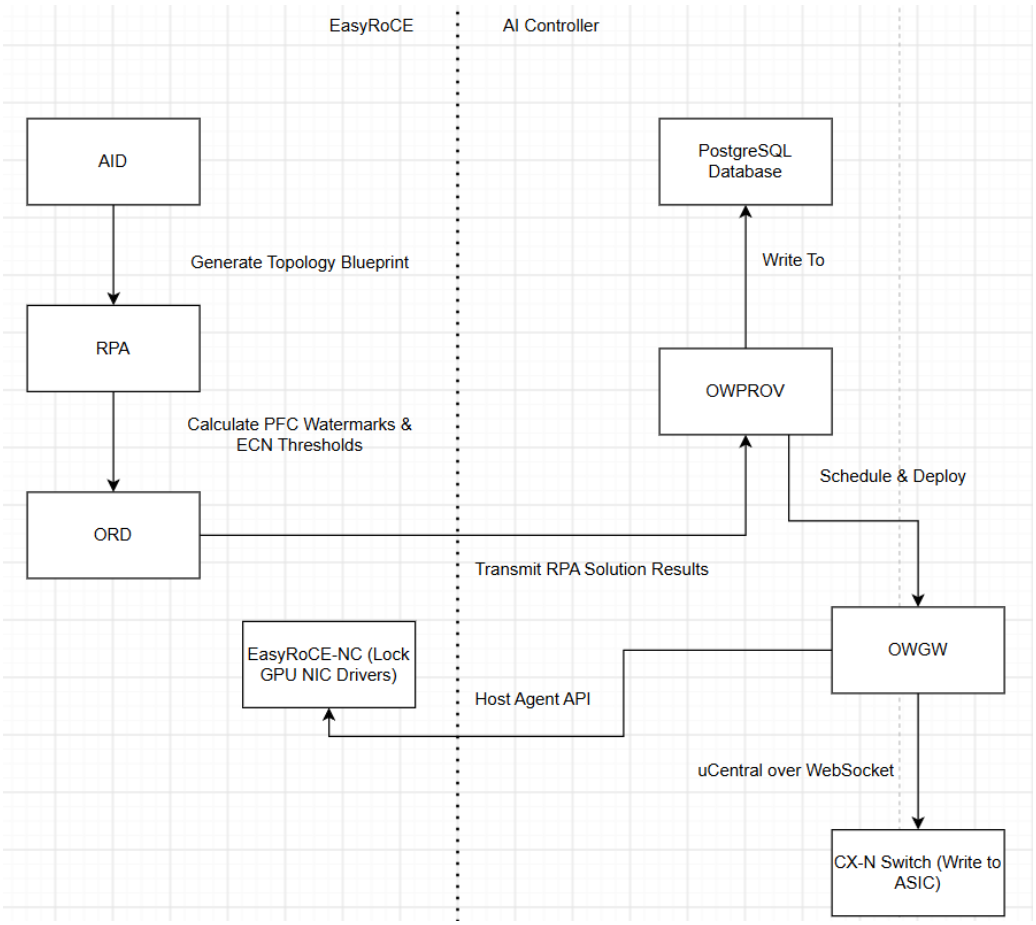


Figure 1: Algorithmic Solving and End-Network Collaborative Deployment Workflow

Mechanism 2: Joint End-Network Telemetry and Slow Worker Self-Healing Loop

In AI training, "Slow Workers" caused by subtle NIC hardware degradation or Hash polarization are notoriously hard to detect. The system eliminates diagnostic blind spots via the following closed loop:

1. **Multi-Source High-Frequency Collection:** Network-side RE/AE exporters streaming switch health and PFC/ECN queue stats in real time, while host-side NE exporter streams physical NIC operational metrics (e.g., RDMA retransmissions, DMA response latency).

2. **High-Throughput Aggregation & Buffering:** Telemetry data is reported to OWGW and concurrently buffered into Kafka message queues asynchronously.

3. **Cross-Correlation Diagnostics (Core):** OWANALYTICS subscribes to Kafka streams, performing timestamp alignment and cross-domain correlation on "Network RE/AE + Host NE" data to determine within seconds whether the root cause is a "NIC hardware defect" or "switch Hash polarization congestion."

4. **Millisecond-Level Closed-Loop Self-Healing:** Upon detecting link congestion, the AI Controller automatically triggers OWGW to issue steering commands, activat-

ing Adaptive Routing Scheme (ARS) on CX-N switches for dynamic flow re-routing around congested paths.

5. **Synergistic Value:** Fuses network and host telemetry inside the controller, removing Slow Worker diagnosis blind spots.

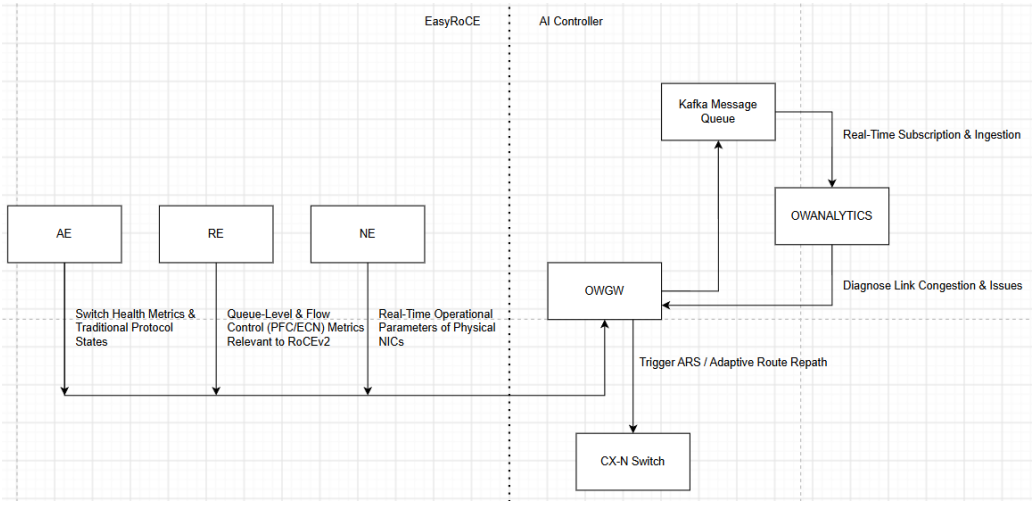


Figure 2: Joint End-Network Telemetry and Slow Worker Self-Healing Workflow

Customer Value Summary and Solution Comparison

Table 1: Traditional Ethernet Solution vs. Asterfusion AI Controller + EasyRoCE Solution

Comparison Dimension	**Traditional Ethernet Solution**	**Asterfusion AI Controller + EasyRoCE Solution**	**Core Customer Value**
Deployment Efficiency	Manual per-device CLI setup taking days; high risk of human error	AID/RPA auto-calculates topology & parameters with one-click dual-path push	Provisioning efficiency boosted by 90%+; eliminates manual configuration errors
End-Network Collaboration	Network tools manage switches, server tools manage NICs; isolated silos	OWGW unified control, forcibly locking network policies and host NIC drivers together	Precise matching of host-network parameters; prevents NIC-network mismatches
Lossless Forwarding	Empirical trial-and-error tuning; prone to PFC storms or packet loss	Algorithmic parameter solving + ASIC microsecond-level hardware self-governance	Guarantees RoCEv2 zero packet loss; maximizes throughput and MFU
Fault Self-Healing	Slow Worker localization takes days; lacks cross-domain data linkage	AE/RE/NE joint telemetry + Kafka + OWANALYTICS + ARS re-routing	Slow Worker localization reduced to seconds; dynamic auto-rerouting bypasses congestion

Summary

Asterfusion's "AI Controller + EasyRoCE" joint end-network collaborative architecture—built upon **automated algorithmic solving**, **synchronized dual-path push**, **hardware-level lossless forwarding**, and **multi-source telemetry self-healing**—systematically resolves the fundamental pain points of traditional AI networks regarding configuration efficiency, tuning complexity, and fault isolation. It provides a simple, lossless, and self-

healing network foundation for ultra-large-scale AI LLM training.